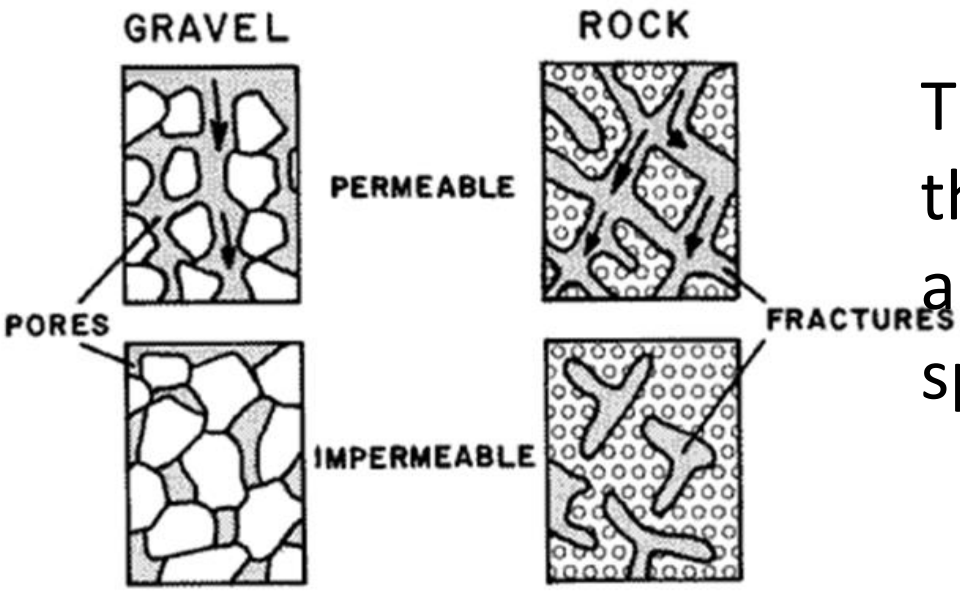
Aquifers

Definition: An aquifer is defined as saturated permeable geological unit that is permeable enough to yield economic quantities of water to wells. The most common aquifer: unconsolidated sand and gravel.

An aquitard is geological unit that is permeable enough to transmit water in significant quantities when viewed over large areas and long periods but its permeability is not sufficient to justify the production of wells being placed in it. Clays, loams, and shales are typical aquitards.

an aquiclude is an impermeable geological unit that does not transmit water at all. Dense unfractured igneous and metamorphic rocks are typical example of it

Aquifers typically consist of gravel, sand, sandstone, or fractured rock such as limestone. These types of materials are permeable because they have large connected spaces that allow water to flow through. The spaces in a gravel aquifer are called pores. The spaces in a fractured rock aquifer are called fractures. If a material contains pores that are not connected, ground water cannot move from one space to another. These materials are said to be impermeable. Materials such as clay or shale have many small pores, but the pores are not well connected. Therefore, clay or shale usually restrict the flow of ground water



The amount of spaces is the porosity. Permeability is a measure of how well the spaces are connected

Aquifers

Confined

(groundwater is confined between layers of clay, dense rock or other materials with very low permeability)

Unconfined

(Do not have a low-permeability deposit above it. The top layer of an unconfined aquifer is the water table.)

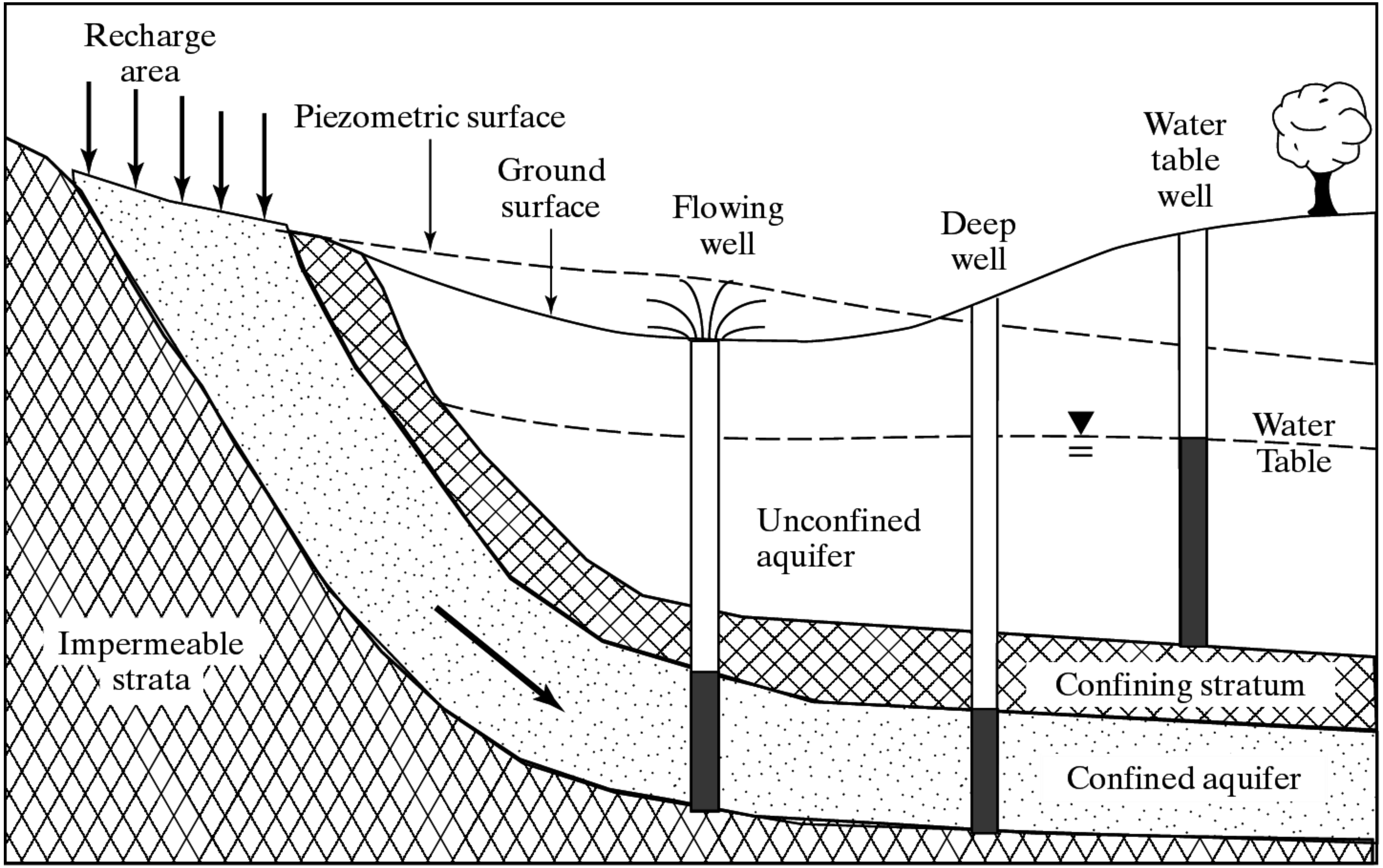
Also called Water Table Aquifer

Semiconfined

(One of the confining layer (Top or bottom) is allowing some leakage because of permeability)

Also called leaky aquifer

Example Layered Aquifer System



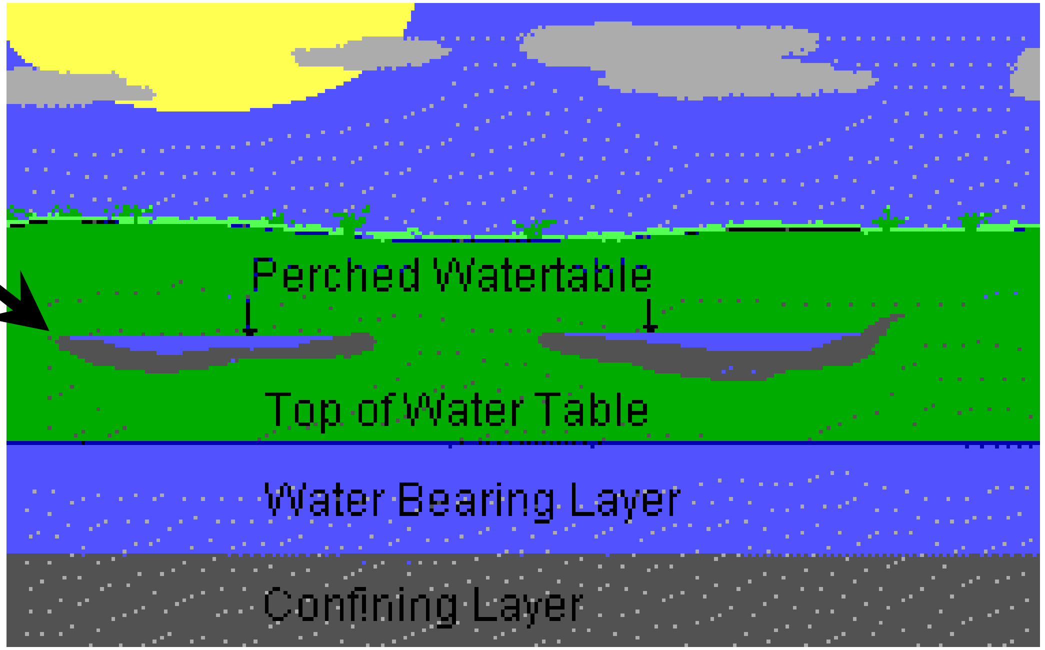
Bedient et al., 1999.

Leaky and Perched Aquifers

- **Leaky confined aquifer: represents a stratum that allows water to flow from above through a leaky confining zone into the underlying aquifer**

- **Perched aquifer: occurs when an unconfined water zone sits on top of a clay lens, separated from the main aquifer below**

In deep sedimentary basins, an interbedded system of permeable and less permeable layers that form a multi-layered aquifer system



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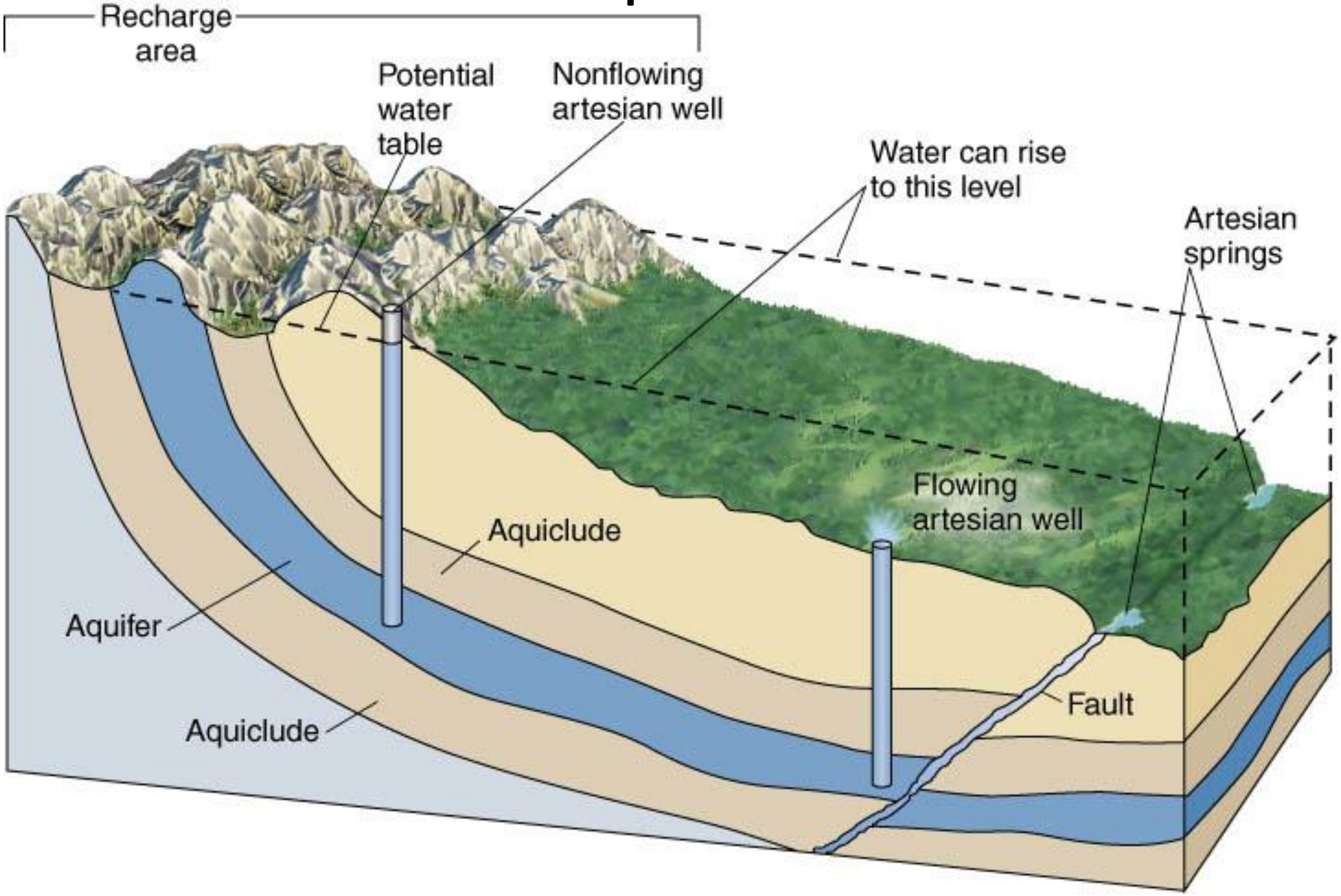
Other Aquifer Features



How to map Depth of Anconfined Aquifer or
Water table

Installation peizometer

The map is created by plotting elevations of the static water level and then generating contours or lines of equal elevation. Static water levels used to develop the potentiometric surface map are from wells completed in aquifer systems at various depths and under confined and unconfined condition

Basic Terms

- **Porosity**
- Permeability
- Bulk Density
- Specific Yield
- Hydraulic Gradient
- Isotropic/Anisotropic
- Homogeneous/Hetrogeneous

Porosity

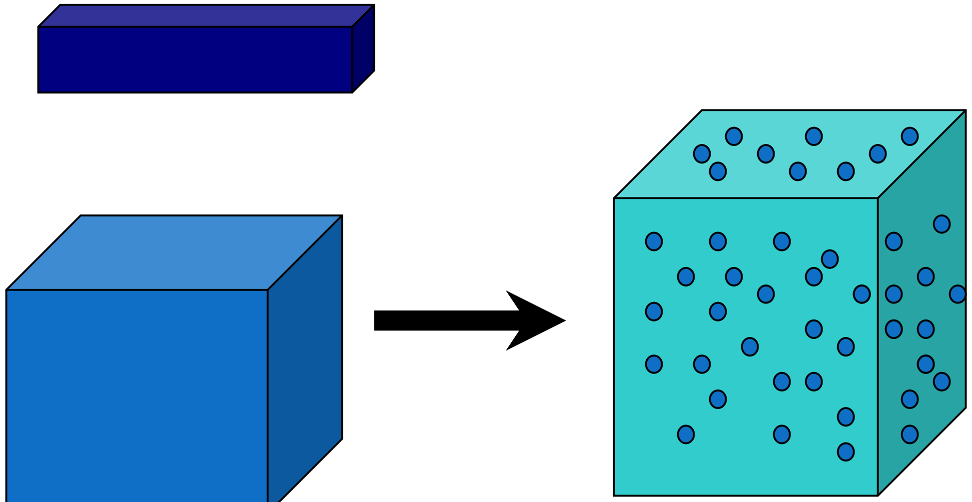
- The porosity of a rock or soil is simply the ratio of volume of voids to total volume of material.

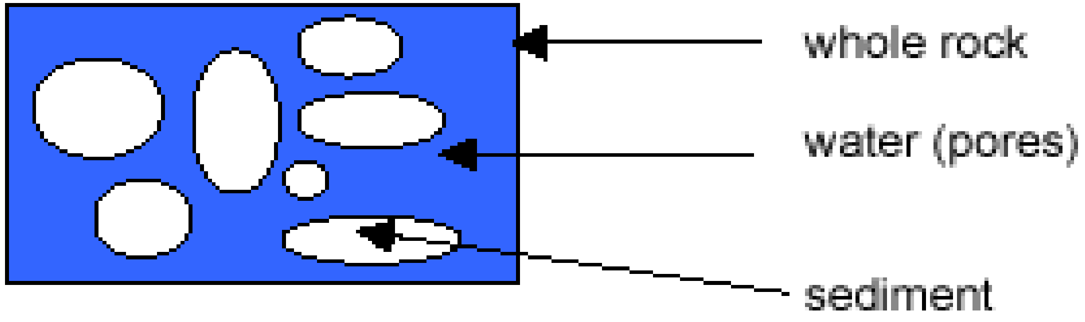
commonly denoted by n $$n = Vv / Vt$$

Vv is the volume of voids in a total volume of material Vt.

Porosity is a dimensionless

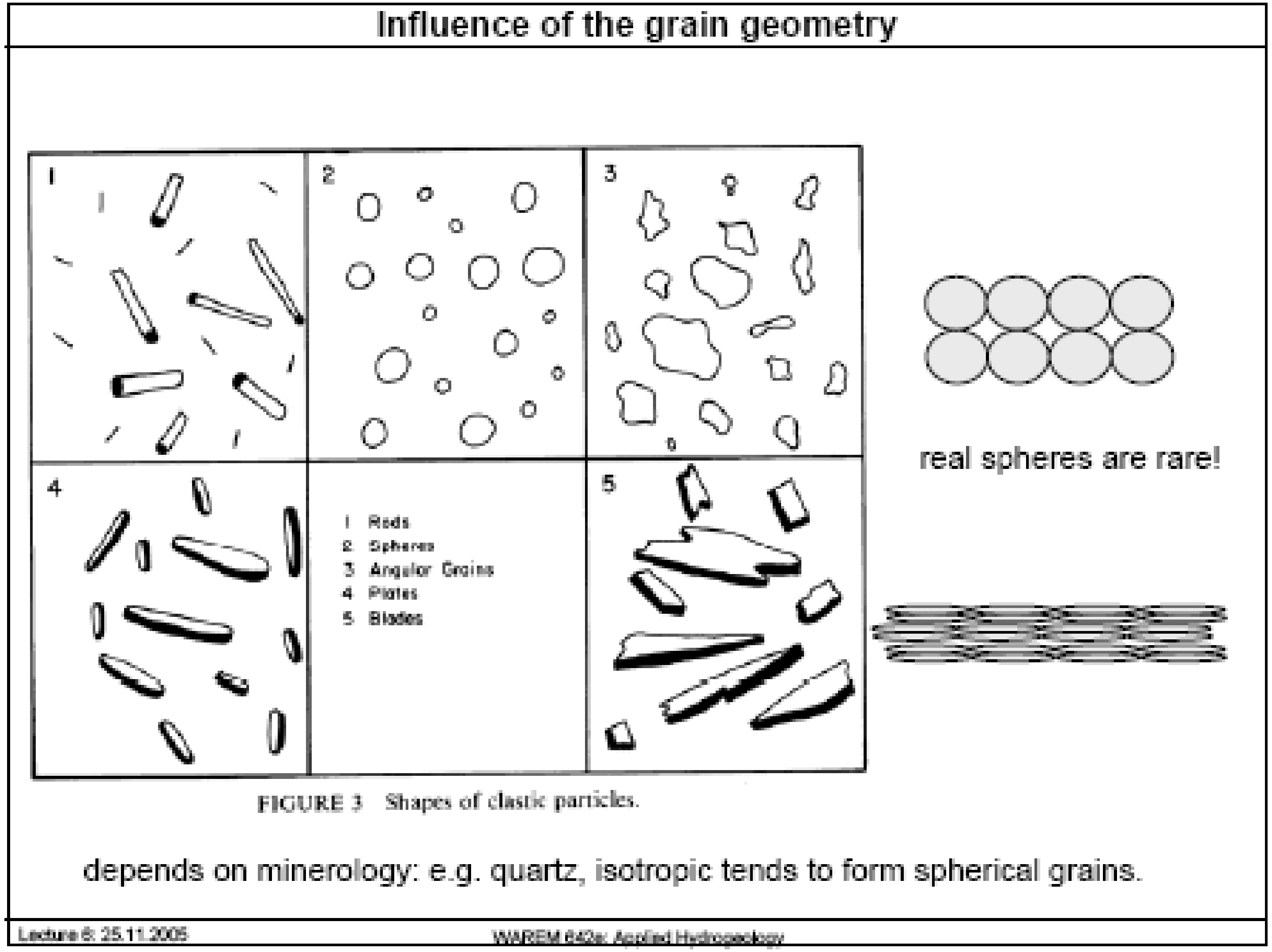
Range $0 < n < 1$.





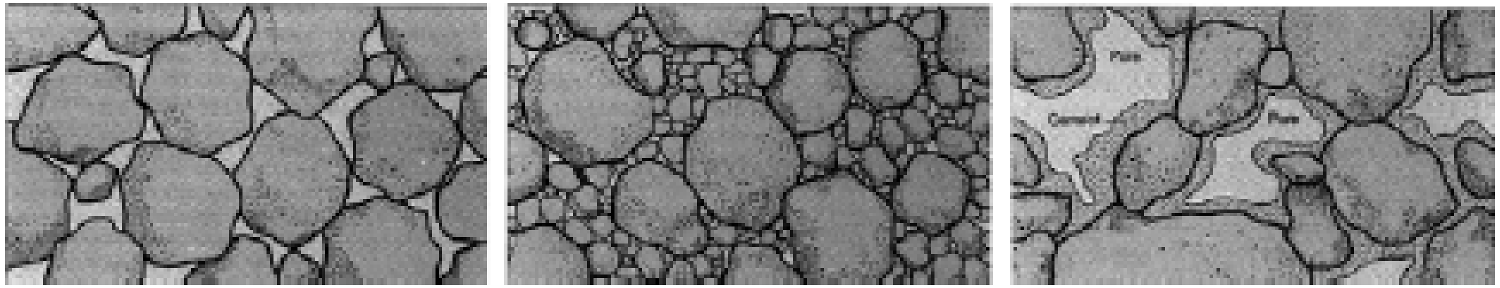
The term effective porosity refers to the amount of interconnected pore space available for fluid flow and is expressed as a ratio of interconnected interstices to total volume

Factors affects Porosity



Cement, grain size distribution

Influence of grain size distribution and cementation on porosity.



Left: Well-sorted sediment with large pore spaces.

Middle: Poorly sorted sediment, pore spaces partly filled with finer grains

Right: Large pores partly filled with cement

Cementation and Recrystallisation

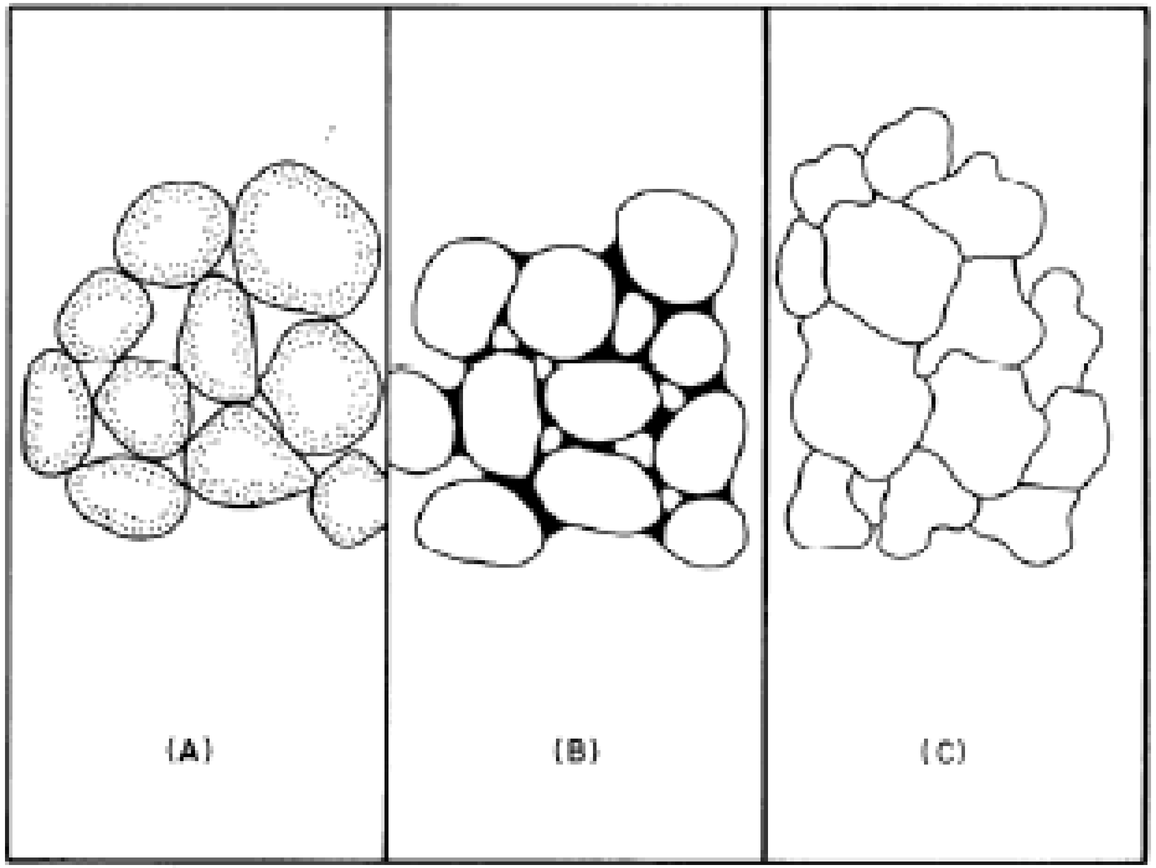


FIGURE 2 Recrystallizing grains.

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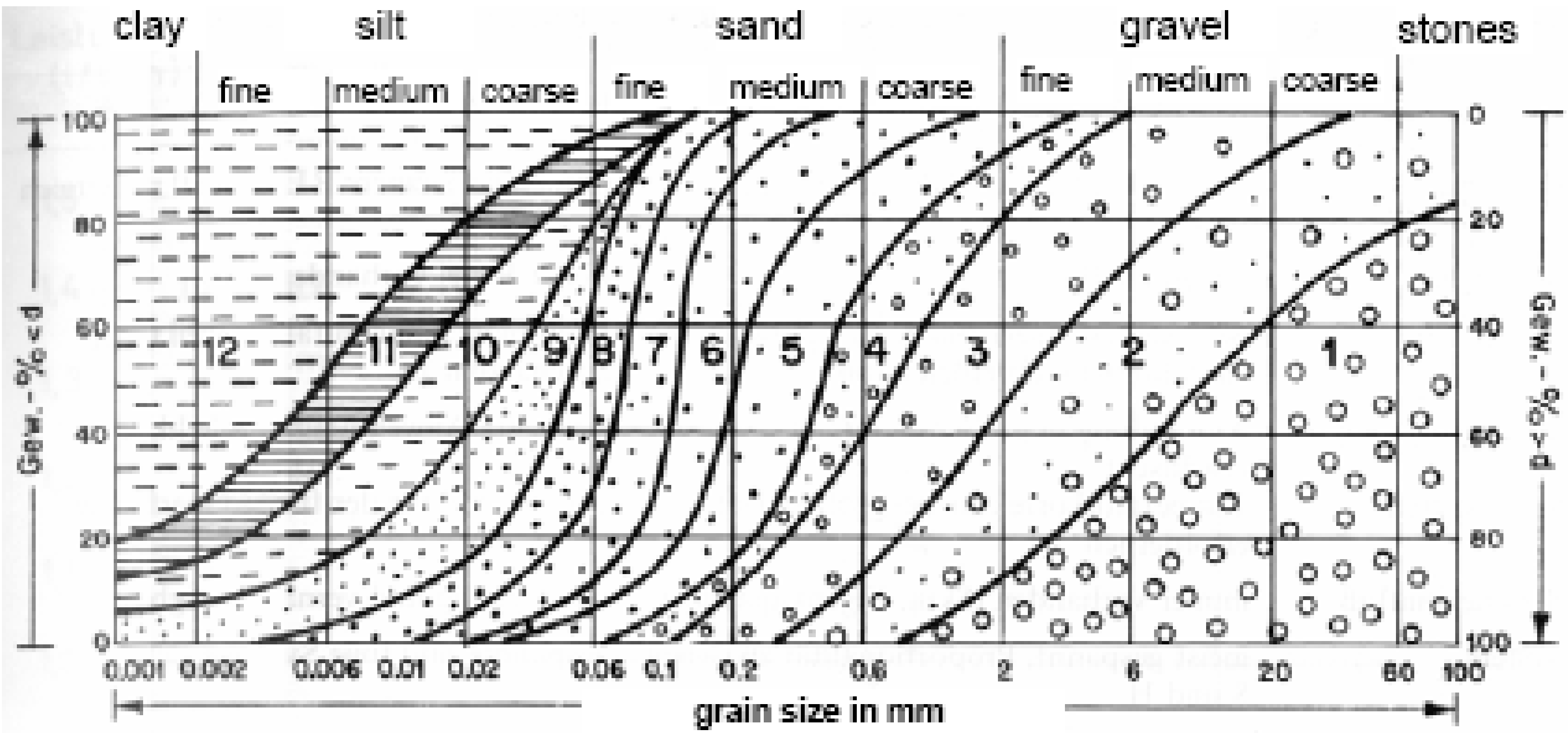
Grain size distribution



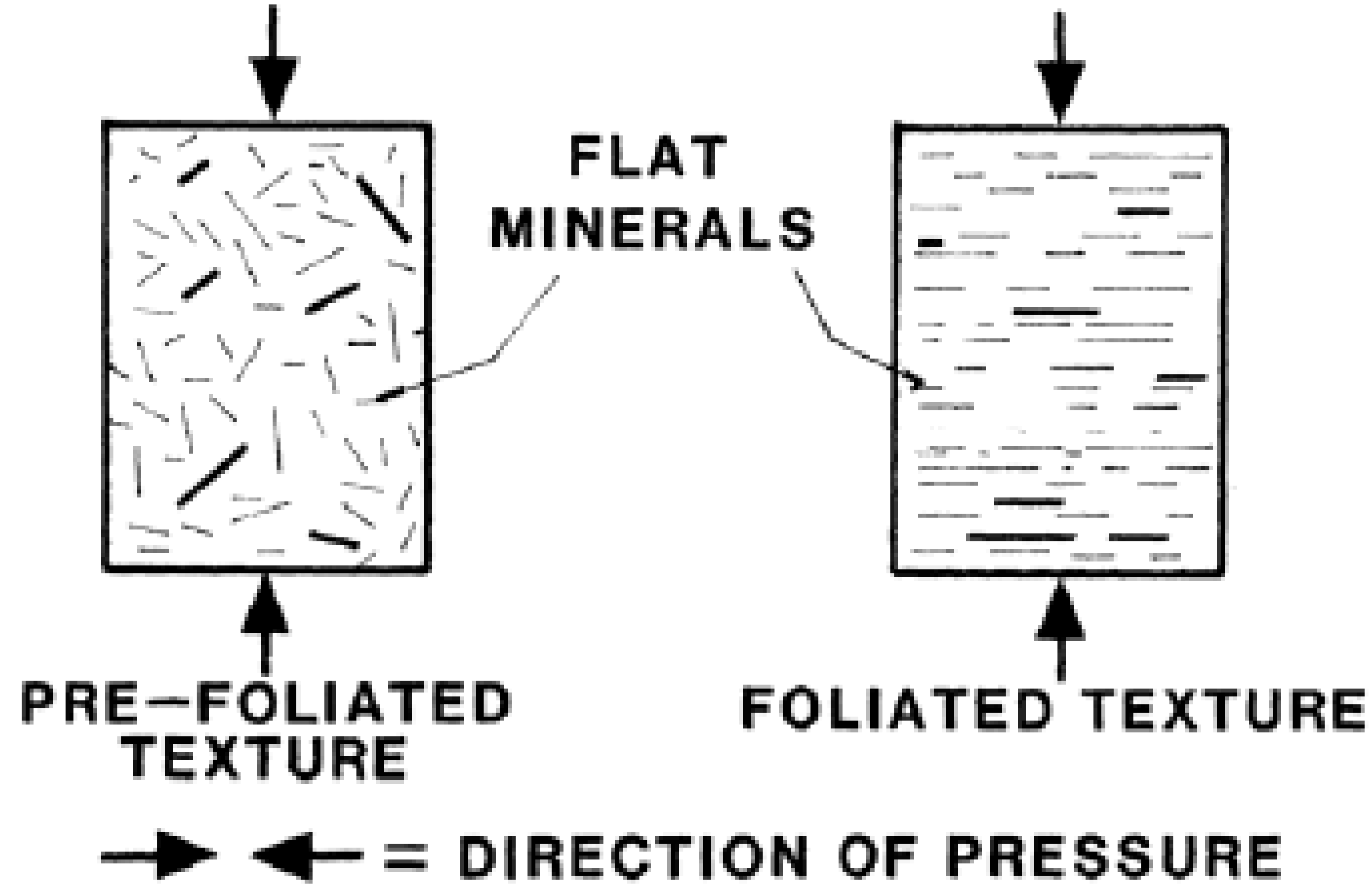
K in m/s (estimates)

12	11	10	9	8	7	6	5	4	3	2	1
$<1\cdot10^{-9}$	$1\cdot10^{-9}$	$1\cdot10^{-8}$	$1\cdot10^{-7}$	$1\cdot10^{-6}$	$1\cdot10^{-5}$	$5\cdot10^{-5}$	$1\cdot10^{-4}$	$5\cdot10^{-4}$	$1\cdot10^{-3}$	$7\cdot10^{-3}$	$3\cdot10^{-2}$

Abb. 1.30. Korngrößenklassen und Durchlässigkeitsbeiwerte (k_f) der Lockergesteine (Prinz 1991).

Balke

Influence of Compaction: Porosity, hydraulic cond., Anisotropy



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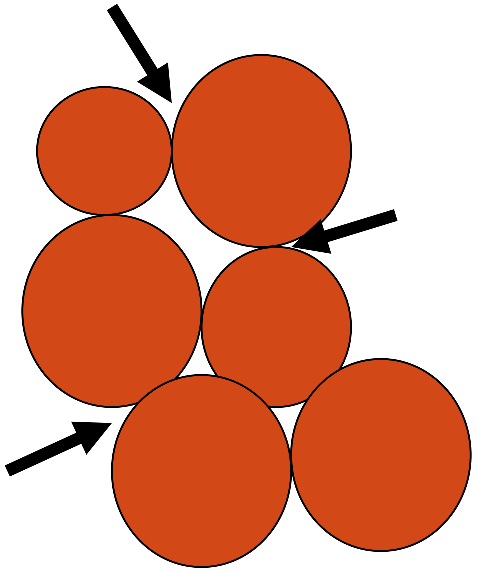
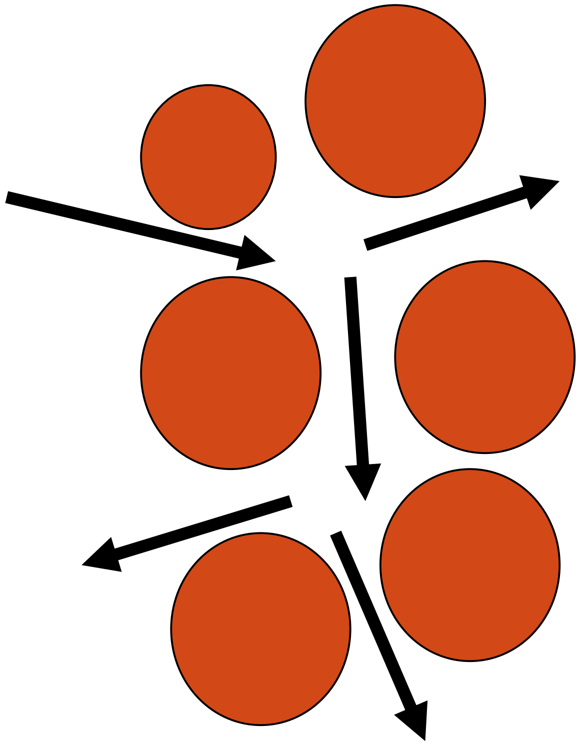
Permeability

- Permeability - If the air spaces within the soil are interconnected, water can flow through the rock and soil and the material is said to be “permeable”

Material can be porous without being permeable, but it cannot be permeable without being porous.



Porous, but not permeable

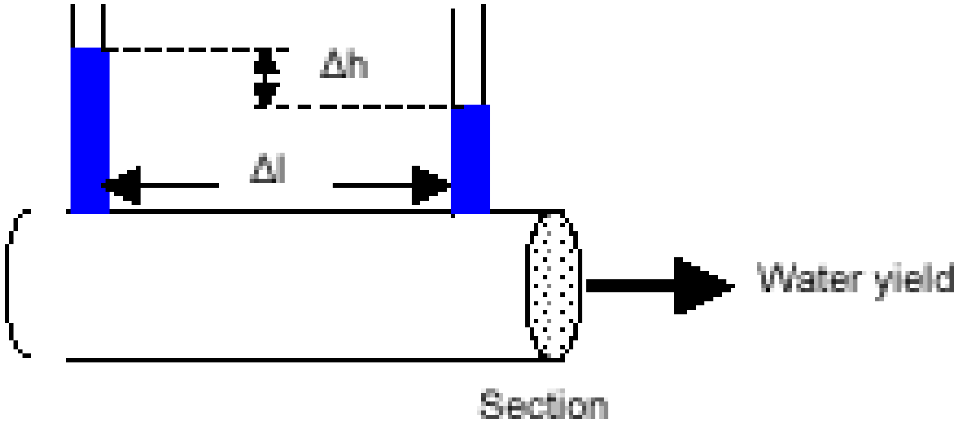


Porous and permeable

PERMEABILITY:

$$\text{Permeability} = \frac{\text{water yield}}{\text{sample section}} \, / \, \text{hydraulic gradient}$$

with hydraulic gradient = Δh / Δl



Zone of Saturation

Effective porosity provide the a direct measurement of the water contained per unit volume, But a portion of water can be removed from surface strata by drainage or by pumping of a well; however molecular and surface tension forces held the remainder of water in place.

Specific Retention (S_r): ***Specific Retention*** is the amount of water that a unit volume of an earth material can retain the pores by molecular attraction and capillarity, after gravity drainage. It is defined mathematically by the equation:

$$S_r = 100\frac{V_r}{V}$$

where, S_r is the specific retention, V_r is the volume of water retained against gravity and V is the bulk volume of the sample.

Since the specific retention represents the volume of water retained in the pores against gravity drainage, it may be evaluated by the following expression,

$$S_r = \theta - S_y$$

Specific Yield (S_y)

Specific Yield (S_y): Specific yield is the quantity of water that a unit volume of unconfined aquifer releases by gravity. It is defined mathematically the the equation

$$S_y = 100\frac{V_g}{V}$$

Where S_y is the specific yield, V_g is the water volume drained by gravity force and V is volume of bulk sample

Specific Yield (S_y) and Specific Retention (S_r)

- Porosity (n) = maximum amount of water that a porous medium can contain when saturated
  - But only part of this water is available to supply a well or a spring
    - The water that will drain under the influence of gravity is called the specific yield (S_y)
      - Is a measure of how much water is available for man's use
    - The water that is retained as a film on the porous medium surfaces and openings is called specific retention (S_r)
      - The forces that control specific retention are the capillary forces
- $n = S_y + S_r$
- $S_y = V_d / V_t$, where:
  - V_d = volume of water drained
  - V_t = volume of porous medium

The specific yield of sediments and rocks is generally determined in situ by aquifer-test or pumping-test methods, such as the Theis-curve procedure or Neuman's semi-logarithmic method. Aquifer-test procedures involve a well from which water is pumped. Water level drawdown is recorded at different times, after pumping is started in one or more nearby observation wells.

Specific yield of sediments and rocks can also be determined by laboratory methods. In this case, one must be sure that the saturated samples remain undisturbed and must be allowed to drain for very long period of time (months) before equilibrium is reached. The ratio of the volume of water drained to the volume of the rock sample is the specific yield. Table..... lists representative specific yield ranges for selected materials.

Sediments	Specific Yield (%)
Clay	1-10
Sand	10-30
Gravel	15-30
Sand and Gravel	15-25
Sandstone	5-15
Shale	0.5-5
Limestone	0.5-5

Specific Storage (S_s): ***Specific storage*** is the amount of water stored or expelled by the compressibility of the mineral skeleton and pore water per unit volume of a confined aquifer and per unit change in head. Specific storage has a dimensions of (1/L) and has value on the order of 10^{-3}.

Storativity or storage coefficient (S) of a confined aquifer is the product of the specific storage (S_s) and the aquifer thickness (b). The storativity of most confined aquifers is between 10^{-3} and 10^{-5}. The storativity for an unconfined aquifer is usually taken to be equal to the specific yield. The specific yield of most alluvial aquifers is between 10 to 30 percent.

Common Values

SELECTED VALUES OF POROSITY, SPECIFIC YIELD, AND SPECIFIC RETENTION

[Values in percent by volume]

Material	Porosity	Specific yield	Specific retention
Soil	55	40	15
Clay	50	2	48
Sand	25	22	3
Gravel	20	19	1
Limestone	20	18	2
Sandstone (semiconsolidated)	11	6	5
Granite	.1	.09	.01
Basalt (young)	11	8	3

Specific yield values are % by volume

Specific Yield

(Volume or % of water yielded by an unit volume of aquifer under gravity)

Total Vol. of water = 100 ml

Bulk volume or gross volume = 1000 ml

Vol. of water withdrawn = 20 ml

Porosity= Vol. of pores/Bulk volume

= 100/1000

= 10%

Sp. Yield = 20/1000

= 2%

Dry bulk density (ρb)

$$\rho b = Ms/Vt$$

where *ms is the mass of solids in sample volume Vt*.

Wet or total bulk density *ρt is defined similarly, but it includes the mass of both solids* and water:

$$\rho t = (Ms + Mw)/Vt$$

where *mw is the mass of water in the sample*.

Water Content

The **volumetric water content** ***θ is the fraction of space occupied by water in a given*** **volume of material** ***Vt****:*

$$\theta = Vw/Vt$$

where *Vw is the volume of water.*

If a material's pores are saturated with water, $\theta = n$;

If the pores contain some air, $\theta < n$

Hydraulic gradient (i) is natural fall in water level with respect to distance taking same datum as reference.

Assuming TWS are having same R.L.

Static water level in TW-1 = 13.15 m (From ground level)

Static water level in TW-2 = 14.05 m (From ground level)

Distance between TW-1 & TW-2 = 50 m

Hydraulic gradient $\Delta h/\Delta l = (14.05-13.15)/50$

$$\mathbf{i = \Delta h/\Delta l = 0.9/50 = 0.018}$$

Isotropy implies that K *is independent of direction at a given location.* $K_x = K_y = K_z$.

Anisotropy implies that the value of K at a given location depends on direction. $K_x \neq K_y \neq K_z$.

In a **heterogeneous** material the value of K *varies spatially, and in a* ***homogeneous*** material K *is independent of location*

Example

Given, $n = 0.30$ unsaturated zone is $\theta u = 0.12$

Calculate Sy for the sand aquifer.

Estimate how much water would be removed from storage in a 1 km2 area, if the head is lowered by 0.5 m

Solution

$$Sy = n - \theta u$$

$$= 0.18$$

The volume of water removed from storage in the unconfined aquifer is

$$dVw = -Sy \text{ x } A \text{ x } dh$$

$$= -(0.18)(1\ )(-0.5\ )$$

$$= 90{,}000 \text{ m}^3$$

Typical Values of Hydraulic Conductivity

Material	K(cm/sec)
Gravel	10^{-1} to 100
Clean sand	10^{-4} to 1
Silty sand	10^{-5} to 10^{-1}
Silt	10^{-7} to 10^{-3}
Glacial till	10^{-10} to 10^{-4}
Clay	10^{-10} to 10^{-6}
Limestone and dolomite	10^{-7} to 1
Fractured basalt	10^{-5} to 1
Sandstone	10^{-8} to 10^{-3}
Igneous and metamorphic rock	10^{-11} to 10^{-2}
Shale	10^{-14} to 10^{-8}

Sources: (1) Freeze and Cherry (1979); (2) Neuzil (1994).